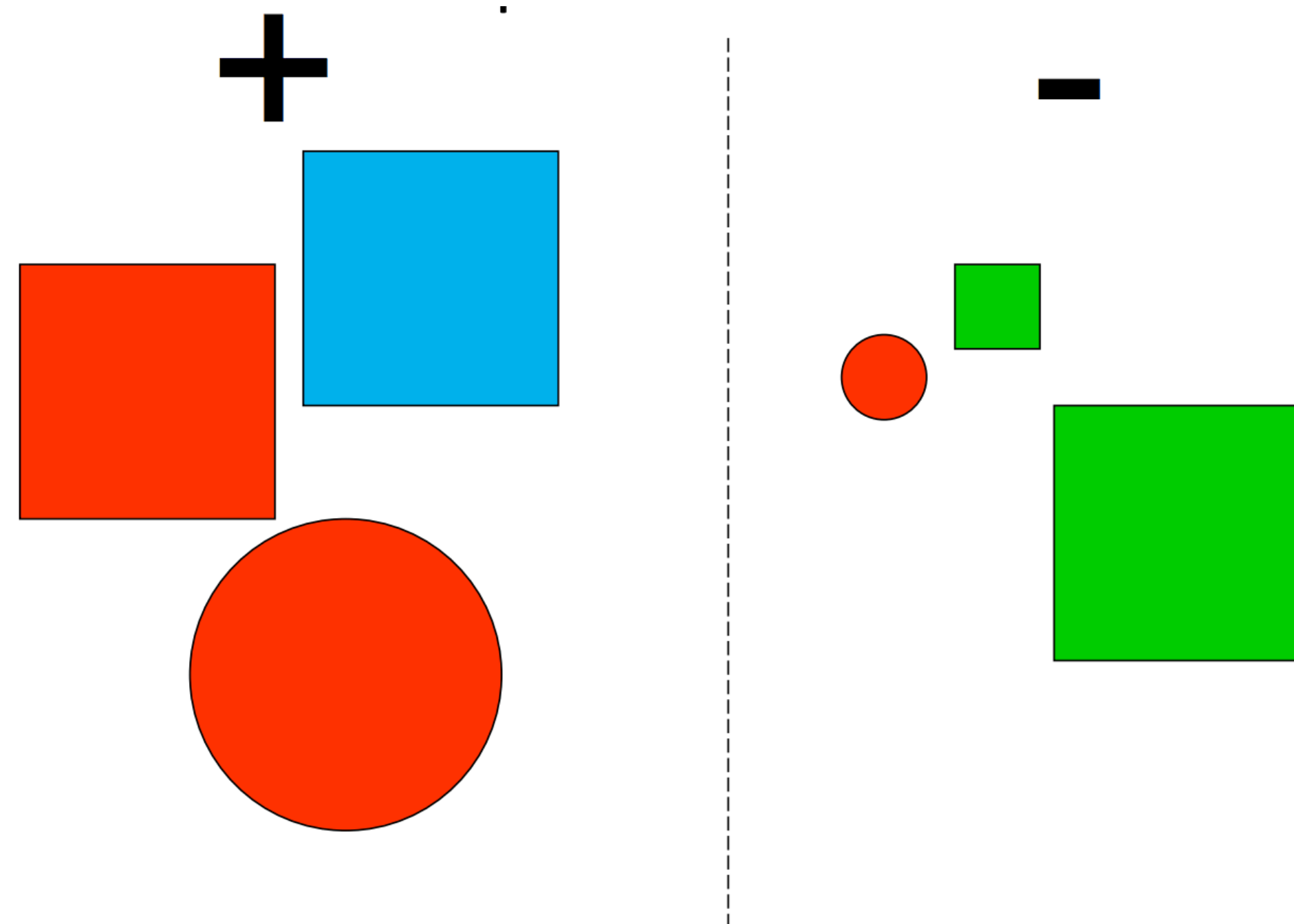


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DATA MINING

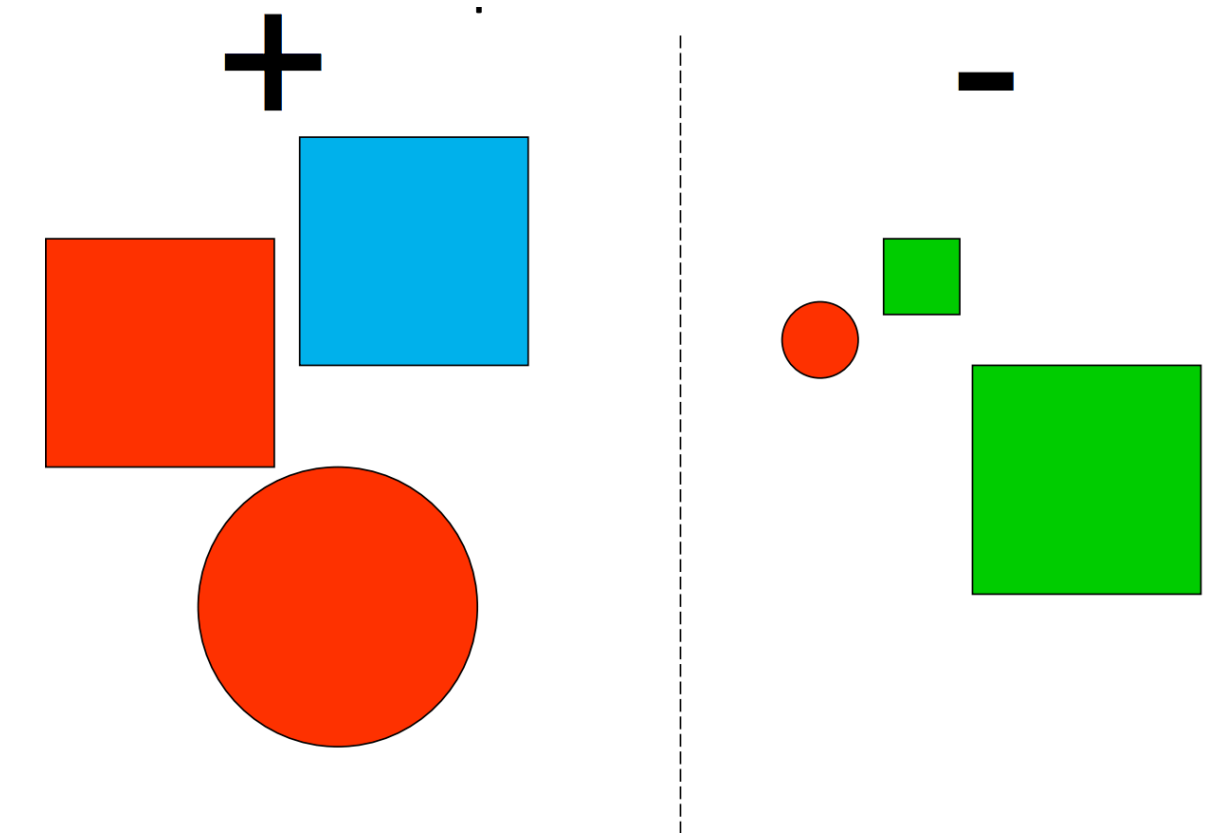
EXAMPLE: CHOOSE THE ATTRIBUTE WITH LARGEST INFORMATION GAIN

- ▶ Features: color, shape, size
- ▶ What's the best feature to use at root?



EXAMPLE: CHOOSE THE ATTRIBUTE WITH LARGEST INFORMATION GAIN

Example	Color	Shape	Size	Class
1	Red	Square	Big	+
2	Blue	Square	Big	+
3	Red	Circle	Big	+
4	Red	Circle	Small	-
5	Green	Square	Small	-
6	Green	Square	Big	-



- ▶ $H(S) = -(0.5 \cdot \log 0.5 + 0.5 \cdot \log 0.5) = 1$
- ▶ $\text{Gain}(S, \text{Color}) = 1 - 0.5 \cdot (-0.67 \cdot \log 0.67 - 0.33 \cdot \log 0.33) - 0.17 \cdot (-1 \cdot \log 1 - 0 \cdot \log 0) - 0.33 \cdot (-1 \cdot \log 1 - 0 \cdot \log 0) = 0.54$
- ▶ $\text{Gain}(S, \text{Shape}) = 1 - 0.67 \cdot (-0.5 \cdot \log 0.5 - 0.5 \cdot \log 0.5) - 0.33 \cdot (-0.5 \cdot \log 0.5 - 0.5 \cdot \log 0.5) = 0$
- ▶ $\text{Gain}(S, \text{Size}) = 1 - 0.67 \cdot (-0.75 \cdot \log 0.75 - 0.25 \cdot \log 0.25) - 0.33 \cdot (-1 \cdot \log 1 - 0 \cdot \log 0) = 0.46$

BUILDING TREE RECURSIVELY

Buildtree(examples, attributes)

*/*examples: a list of training examples at the current node
attributes: a set of candidate attributes to place question on*/*

If examples={} **then** return

If examples have the same label y **then** return a leaf node with label y

If attributes={} **then** return a leaf node with the majority label in examples

$A = \text{Best_attribute}(\text{examples}, \text{attributes})$ */*Suppose attribute A has n possible values*/*

Create an internal node, $\text{node}(A)$, with n children

For attribute A 's i -th possible value $A(i)$:

The i -th child of $\text{node}(A) = \text{Buildtree}(\{\text{examples with its value on } A \text{ being } A(i)\}, \text{attributes}-\{A\})$

PREDICTIVE MODELING

DEALING WITH CONTINUOUS ATTRIBUTES

- ▶ Discretize the value of a continuous attribute into several intervals
 - ▶ Two bins for the continuous variable x : $x \leq t$, $x > t$
 - ▶ $\text{Gain}(S, x, t) = \text{Entropy}(S) - |S_{x \leq t}| * \text{Entropy}(S_{x \leq t}) / |S| - |S_{x > t}| * \text{Entropy}(S_{x > t}) / |S|$
 - ▶ $\text{Gain}(S, x) = \max_t \text{Gain}(S, x, t)$

ADDITIONAL ATTRIBUTE SELECTION CRITERIA: GINI GAIN

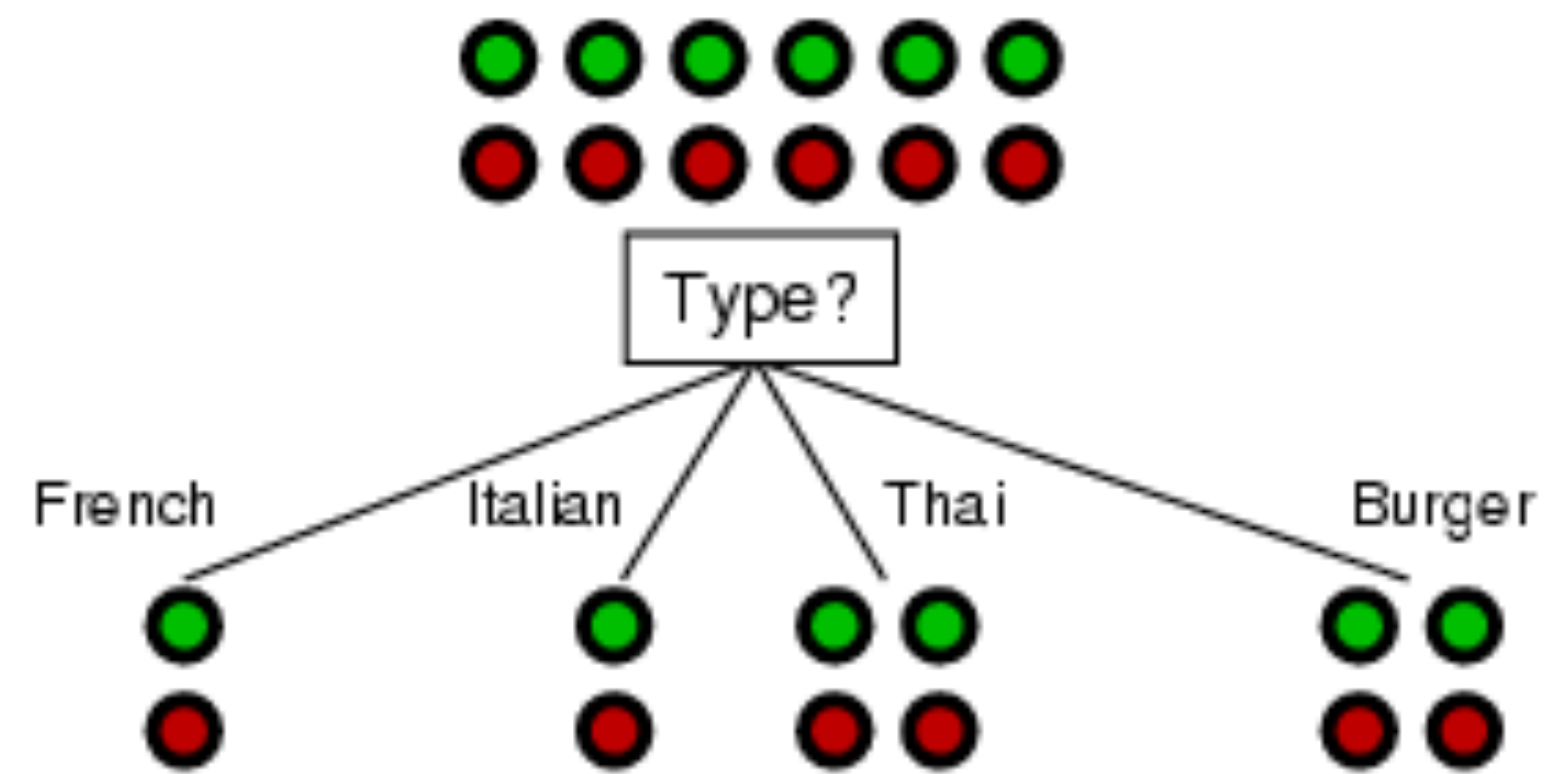
- ▶ Similar to information gain
- ▶ Uses gini index instead of entropy

$$Gini(X) = 1 - \sum_x p(x)^2$$

- ▶ Measures decrease in gini index after split:

$$Gain(S, A) = Gini(S) - \sum_{v \in values(A)} \frac{|S_A|}{|S|} Gini(S_A)$$

COMPARING INFORMATION GAIN TO GINI GAIN

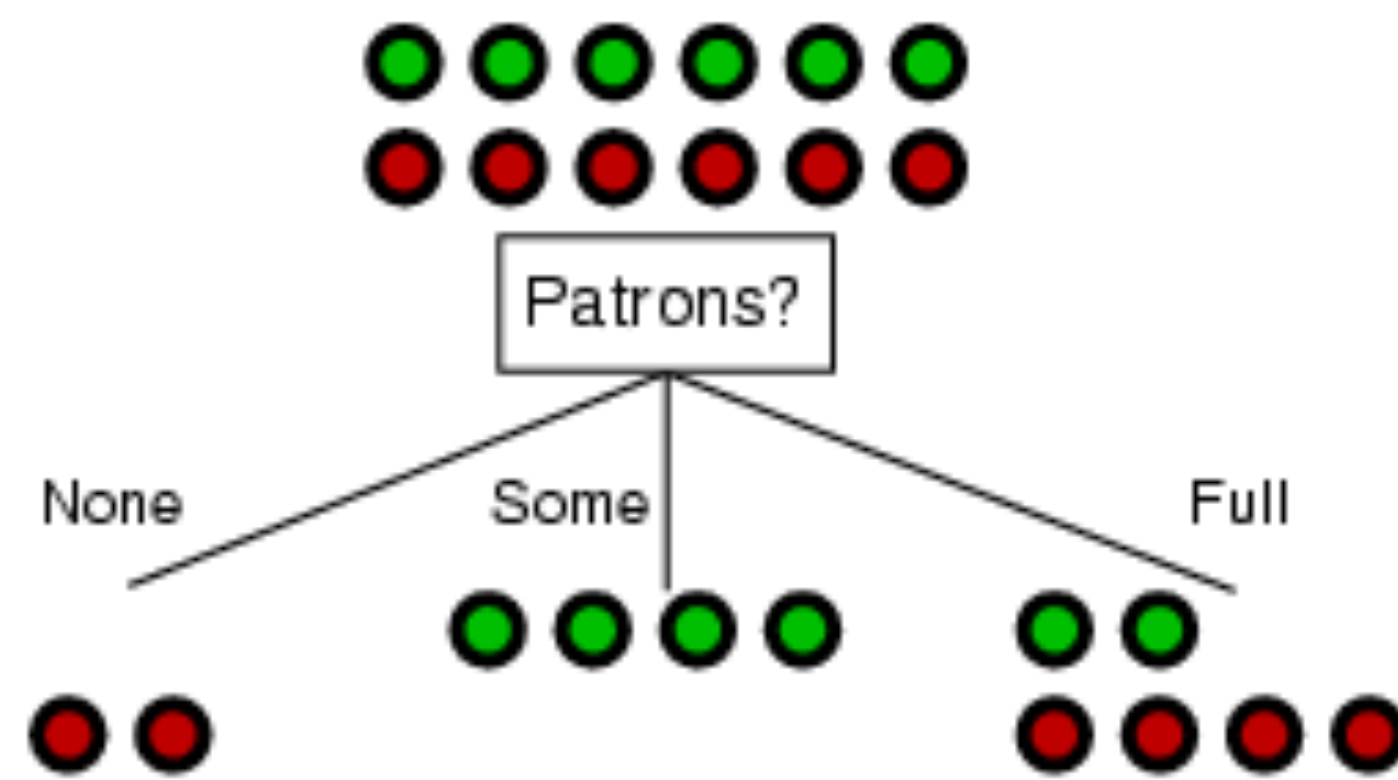


$$IG = 0$$

$$GG = 0$$



COMPARING INFORMATION GAIN TO GINI GAIN



$$IG = 1.0 - \left[\frac{2}{12} 0 \right] - \left[\frac{4}{12} 0 \right] - \left[\frac{6}{12} 0.919 \right] = 0.541$$

$$GG = 0.5 - \left[\frac{2}{12} 0 \right] - \left[\frac{4}{12} 0 \right] - \left[\frac{6}{12} 0.444 \right] = 0.278$$

ADDITIONAL ATTRIBUTE SELECTION CRITERIA: CHI-SQUARE SCORE

- ▶ Widely used to test independence between two categorical attributes (e.g., feature and class label)
- ▶ Considers counts in a contingency table and calculates the normalized squared deviation of observed (actual) values from expected (predicted) values

$$\chi^2 = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}$$

- ▶ Sampling distribution is known to be chi-square distributed, given that cell counts are above minimum thresholds

CALCULATING EXPECTED VALUES FOR A CELL

$$\chi^2 = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}$$

$$o_{(0,+)} = a$$

$$e_{(0,+)} = p(A = 0, C = +) \cdot N$$

$$= p(A = 0)p(C = +|A = 0) \cdot N$$

$$= p(A = 0)p(C = +) \cdot N$$

$$= \left[\frac{a + b}{N} \right] \cdot \left[\frac{a + c}{N} \right] \cdot N$$

Class

	+	-
0	a	b
1	c	d

Attribute

N

(assuming independence)

EXAMPLE CALCULATION

	Observed			Expected	
	Buy	No buy		Buy	No buy
High	2	2	High	2.57	1.43
Med	4	2	Med	3.86	2.14
Low	3	1	Low	2.57	1.43

$$\begin{aligned} \chi^2 &= \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i} = \left(\frac{(2 - 2.57)^2}{2.57} \right) + \left(\frac{(4 - 3.86)^2}{3.86} \right) + \left(\frac{(3 - 2.57)^2}{2.57} \right) \\ &\quad + \left(\frac{(2 - 1.43)^2}{1.43} \right) + \left(\frac{(2 - 2.14)^2}{2.14} \right) + \left(\frac{(1 - 1.43)^2}{1.43} \right) \\ &= 0.57 \end{aligned}$$

PREDICTIVE MODELING

WHEN TO STOP GROWING

- ▶ Full growth methods
 - ▶ There are no examples left
 - ▶ All examples at a node belong to the same class
 - ▶ There are no attributes left for further splits
- ▶ What impact does this have on the quality of the learned trees?
 - ▶ Trees **overfit** the training data and accuracy on testing data suffers

OVERFITTING

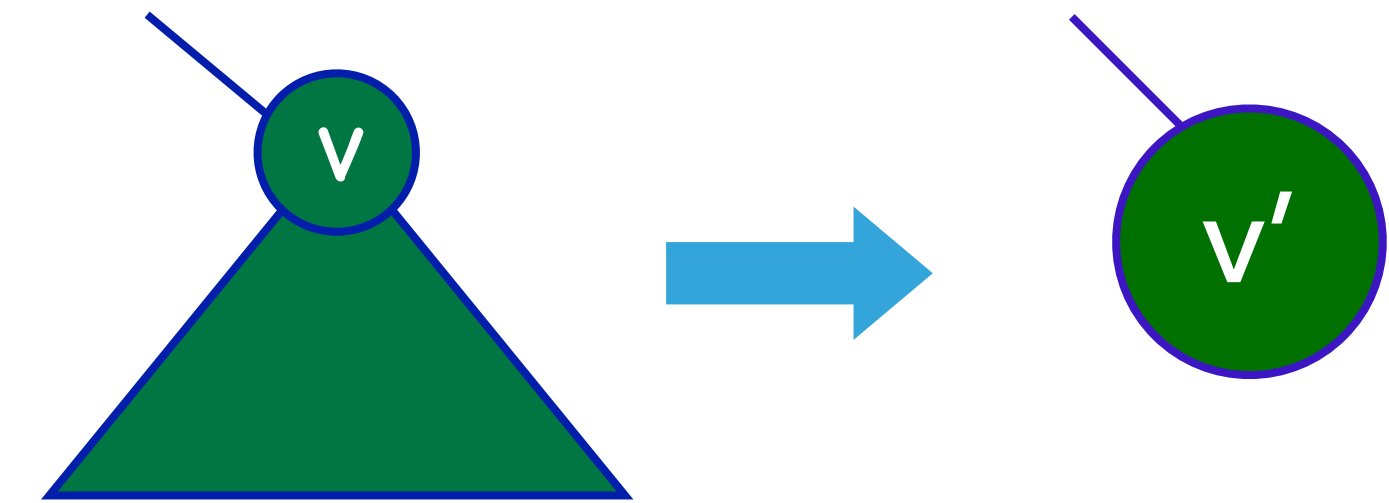
- ▶ Consider a distribution D of data representing a population and a sample D_S drawn from D , which is used as training data
- ▶ Given a model space M , a score function S , and a learning algorithm that returns a model $m \in M$, the algorithm **overfits** the training data D_S if:
 $\exists m' \in M$ such that $S(m, D_S) > S(m', D_S)$ but $S(m, D) < S(m', D)$
- ▶ In other words, there is another model (m') that is better on the entire distribution and if we had learned from the full data we would have selected it instead

HOW TO AVOID OVERFITTING IN DECISION TREES

- ▶ Post-pruning
 - ▶ Separate the training data into a training set and a validation set (i.e., a pruning set).
 - ▶ Fully grow a tree
 - ▶ Use the pruning set to evaluate the utility of pruning (i.e. deleting) nodes from the tree
- ▶ Pre-pruning
 - ▶ Apply a statistical test to decide whether to expand a node
 - ▶ Add penalty terms in scoring functions to prefer trees with smaller sizes

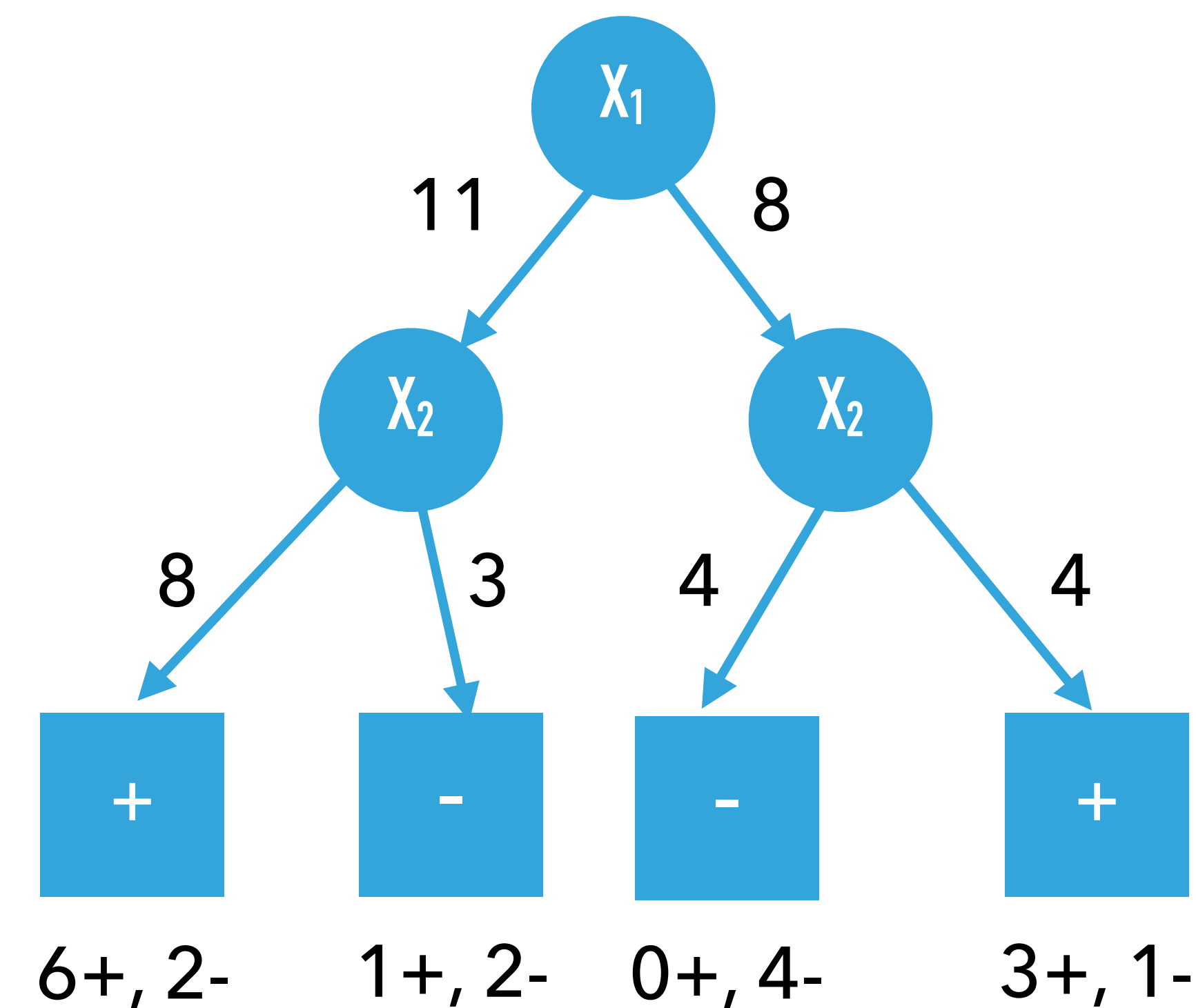
POST-PRUNING METHOD: REDUCED ERROR PRUNING

- ▶ Grow a full tree T using the **training set**
- ▶ Let v be an internal node of the current tree T
- ▶ If we prune at v to create a new tree T' , in T' , the subtree rooted at v will be replaced by a leaf node v' , whose label is the majority label for all **training examples** fall under v
- ▶ Define the gain of pruning at v as, in the **pruning set**, # of misclassified examples under v (in T) - # of misclassified examples that in v' (in T')
- ▶ Repeat: Prune at node with largest gain until only negative gain nodes remain



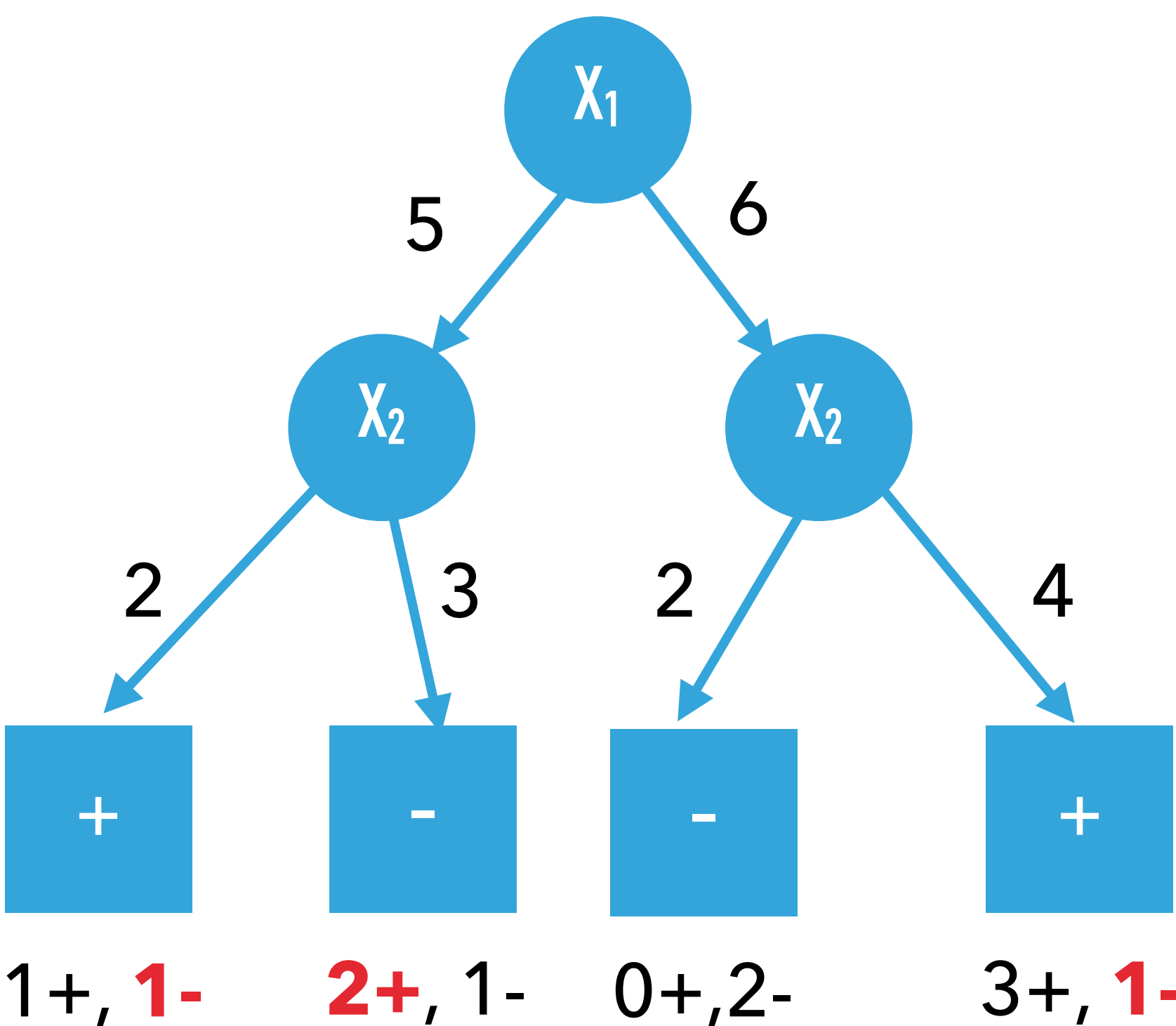
REDUCED ERROR PRUNING EXAMPLE

Training set



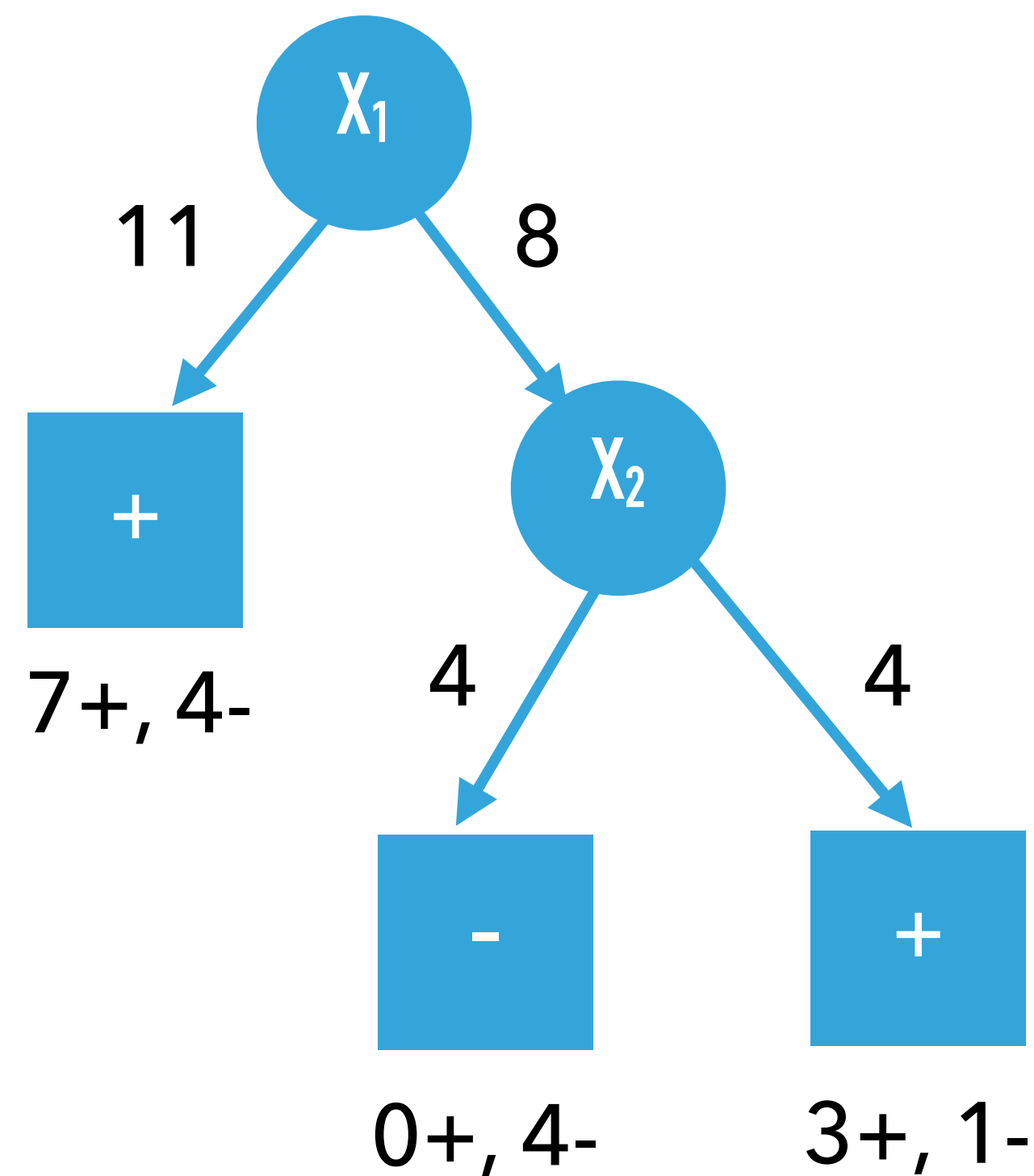
Gain at node X_2 (left):
 $3-2=1$
Gain at node X_2 (right):
 $1-3=-2$
Gain at node X_1 : $4-5=-1$
Prune at node X_2 (left)!

Pruning set



REDUCED ERROR PRUNING EXAMPLE

Training set



Gain at node X_2 : $1-3=-2$

Gain at node X_1 : $3-5=-2$

Done!

Pruning set

